

**BALLENAS ISLAND, BC, Canada**

WMO: 717690

Lat: **49.3506N**Lon: **124.1606W**Elev: **10**StdP: **101.20**Time Zone: **-8.00 (NAP)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -0.7       | 0.7 | -7.2                          | 2.0 | 0.8  | -4.9 | 2.5 | 2.3  | 16.9                  | 4.4  | 15.8 | 5.9  | 4.7                   | 230  | 0.659 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 4.6                    | 23.5            | 18.8 | 22.3 | 18.3 | 21.3 | 17.7 | 19.7                | 22.5 | 18.8 | 21.5 | 18.1 | 20.8 | 4.7                  | 280  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.6                            | 13.4 | 21.6 | 17.7 | 12.7 | 20.6 | 16.9 | 12.1 | 19.9 | 56.3          | 22.6 | 53.5 | 21.6 | 51.3 | 20.9 | 23.7              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |             |             | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |             |             |             |             |             |             |  |  |
|-------------------|-------------|-------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|--|--|
|                   |             |             | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years  |             | n=20 years  |             | n=50 years  |             |  |  |
| 1%                | 2.5%        | 5%          | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min         | Max         | Min         | Max         | Min         | Max         |  |  |
| (a)               | (b)         | (c)         | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)         | (l)         | (m)         | (n)         | (o)         | (p)         |  |  |
| (4) <b>15.9</b>   | <b>13.7</b> | <b>12.1</b> | <b>-2.2</b>                | <b>26.4</b> | <b>1.6</b>         | <b>1.7</b> | <b>-3.4</b>                                        | <b>27.6</b> | <b>-4.4</b> | <b>28.6</b> | <b>-5.3</b> | <b>29.6</b> | <b>-6.5</b> | <b>30.9</b> |  |  |
| (5)               |             |             | <b>-4.1</b>                | <b>20.4</b> | <b>1.4</b>         | <b>1.2</b> | <b>-5.1</b>                                        | <b>21.3</b> | <b>-5.9</b> | <b>22.0</b> | <b>-6.7</b> | <b>22.6</b> | <b>-7.7</b> | <b>23.5</b> |  |  |

**Monthly Climatic Design Conditions**

|                                                                               |             | Annual       | Jan          | Feb          | Mar          | Apr          | May          | Jun          | Jul          | Aug          | Sep          | Oct          | Nov          | Dec         |
|-------------------------------------------------------------------------------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|-------------|
|                                                                               |             | (d)          | (e)          | (f)          | (g)          | (h)          | (i)          | (j)          | (k)          | (l)          | (m)          | (n)          | (o)          | (p)         |
| (6) <b>Temperatures, Degree-Days and Degree-Hours</b>                         | DBAvg       | <b>11.3</b>  | <b>5.9</b>   | <b>5.9</b>   | <b>7.4</b>   | <b>9.7</b>   | <b>13.3</b>  | <b>15.9</b>  | <b>18.4</b>  | <b>18.4</b>  | <b>15.4</b>  | <b>11.3</b>  | <b>7.8</b>   | <b>6.0</b>  |
|                                                                               | DBStd       | <b>5.16</b>  | <b>2.64</b>  | <b>2.28</b>  | <b>1.99</b>  | <b>2.01</b>  | <b>2.36</b>  | <b>2.32</b>  | <b>1.94</b>  | <b>1.89</b>  | <b>2.00</b>  | <b>1.95</b>  | <b>2.49</b>  | <b>2.35</b> |
|                                                                               | HDD10.0     | <b>564</b>   | <b>127</b>   | <b>116</b>   | <b>82</b>    | <b>29</b>    | <b>2</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>9</b>     | <b>72</b>    | <b>126</b>  |
|                                                                               | HDD18.3     | <b>2622</b>  | <b>385</b>   | <b>349</b>   | <b>338</b>   | <b>259</b>   | <b>156</b>   | <b>81</b>    | <b>22</b>    | <b>22</b>    | <b>89</b>    | <b>219</b>   | <b>316</b>   | <b>384</b>  |
|                                                                               | CDD10.0     | <b>1044</b>  | <b>1</b>     | <b>0</b>     | <b>2</b>     | <b>19</b>    | <b>105</b>   | <b>177</b>   | <b>261</b>   | <b>261</b>   | <b>163</b>   | <b>48</b>    | <b>6</b>     | <b>1</b>    |
|                                                                               | CDD18.3     | <b>61</b>    | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>1</b>     | <b>8</b>     | <b>25</b>    | <b>25</b>    | <b>2</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>    |
|                                                                               | CDH23.3     | <b>44</b>    | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>8</b>     | <b>21</b>    | <b>15</b>    | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>    |
|                                                                               | CDH26.7     | <b>2</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>1</b>     | <b>1</b>     | <b>1</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>    |
| (14) <b>Wind</b>                                                              | WSAvg       | <b>5.6</b>   | <b>6.1</b>   | <b>5.5</b>   | <b>5.7</b>   | <b>5.3</b>   | <b>5.0</b>   | <b>5.0</b>   | <b>5.3</b>   | <b>5.0</b>   | <b>5.1</b>   | <b>5.8</b>   | <b>6.4</b>   | <b>6.6</b>  |
| (15) <b>Precipitation</b>                                                     | PrecAvg     | <b>886</b>   | <b>130</b>   | <b>75</b>    | <b>82</b>    | <b>52</b>    | <b>42</b>    | <b>43</b>    | <b>24</b>    | <b>26</b>    | <b>41</b>    | <b>102</b>   | <b>127</b>   | <b>144</b>  |
|                                                                               | PrecMax     | <b>1089</b>  | <b>251</b>   | <b>155</b>   | <b>165</b>   | <b>101</b>   | <b>89</b>    | <b>80</b>    | <b>46</b>    | <b>97</b>    | <b>115</b>   | <b>223</b>   | <b>289</b>   | <b>206</b>  |
|                                                                               | PrecMin     | <b>592</b>   | <b>57</b>    | <b>2</b>     | <b>19</b>    | <b>21</b>    | <b>11</b>    | <b>12</b>    | <b>6</b>     | <b>3</b>     | <b>4</b>     | <b>33</b>    | <b>39</b>    | <b>50</b>   |
|                                                                               | PrecStd     | <b>144</b>   | <b>51</b>    | <b>36</b>    | <b>35</b>    | <b>22</b>    | <b>19</b>    | <b>21</b>    | <b>12</b>    | <b>20</b>    | <b>29</b>    | <b>47</b>    | <b>63</b>    | <b>45</b>   |
| (19) <b>Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures</b> | 0.4%        | DB           | <b>12.1</b>  | <b>11.0</b>  | <b>13.1</b>  | <b>17.1</b>  | <b>21.1</b>  | <b>24.3</b>  | <b>25.5</b>  | <b>25.0</b>  | <b>22.0</b>  | <b>17.5</b>  | <b>14.1</b>  | <b>12.2</b> |
|                                                                               |             | MCWB         | <b>9.6</b>   | <b>8.9</b>   | <b>10.2</b>  | <b>13.0</b>  | <b>15.5</b>  | <b>18.0</b>  | <b>20.3</b>  | <b>19.5</b>  | <b>17.7</b>  | <b>15.7</b>  | <b>12.5</b>  | <b>10.4</b> |
|                                                                               | 2%          | DB           | <b>10.5</b>  | <b>9.8</b>   | <b>11.7</b>  | <b>15.1</b>  | <b>19.0</b>  | <b>22.2</b>  | <b>23.6</b>  | <b>23.4</b>  | <b>20.3</b>  | <b>15.6</b>  | <b>12.4</b>  | <b>10.6</b> |
|                                                                               |             | MCWB         | <b>8.9</b>   | <b>8.2</b>   | <b>9.5</b>   | <b>12.1</b>  | <b>14.9</b>  | <b>17.6</b>  | <b>19.1</b>  | <b>18.9</b>  | <b>17.2</b>  | <b>14.2</b>  | <b>11.3</b>  | <b>9.2</b>  |
|                                                                               | 5%          | DB           | <b>9.6</b>   | <b>9.1</b>   | <b>10.7</b>  | <b>13.6</b>  | <b>17.8</b>  | <b>20.6</b>  | <b>22.4</b>  | <b>22.2</b>  | <b>19.2</b>  | <b>14.6</b>  | <b>11.7</b>  | <b>9.7</b>  |
|                                                                               |             | MCWB         | <b>8.5</b>   | <b>7.8</b>   | <b>8.8</b>   | <b>11.0</b>  | <b>14.3</b>  | <b>16.7</b>  | <b>18.5</b>  | <b>18.5</b>  | <b>16.5</b>  | <b>13.3</b>  | <b>10.8</b>  | <b>8.6</b>  |
|                                                                               | 10%         | DB           | <b>9.0</b>   | <b>8.6</b>   | <b>10.0</b>  | <b>12.5</b>  | <b>16.6</b>  | <b>19.1</b>  | <b>21.3</b>  | <b>21.2</b>  | <b>18.2</b>  | <b>13.8</b>  | <b>10.9</b>  | <b>9.0</b>  |
|                                                                               |             | MCWB         | <b>8.1</b>   | <b>7.4</b>   | <b>8.3</b>   | <b>10.2</b>  | <b>13.7</b>  | <b>15.8</b>  | <b>17.8</b>  | <b>17.9</b>  | <b>15.9</b>  | <b>12.5</b>  | <b>10.1</b>  | <b>8.0</b>  |
| (27) <b>Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures</b> | 0.4%        | WB           | <b>10.4</b>  | <b>9.6</b>   | <b>11.2</b>  | <b>13.8</b>  | <b>16.6</b>  | <b>19.4</b>  | <b>21.1</b>  | <b>21.3</b>  | <b>18.7</b>  | <b>16.3</b>  | <b>13.0</b>  | <b>11.1</b> |
|                                                                               |             | MCDB         | <b>11.0</b>  | <b>10.1</b>  | <b>12.4</b>  | <b>16.2</b>  | <b>19.4</b>  | <b>22.7</b>  | <b>24.6</b>  | <b>23.5</b>  | <b>20.8</b>  | <b>17.1</b>  | <b>13.8</b>  | <b>11.7</b> |
|                                                                               | 2%          | WB           | <b>9.5</b>   | <b>8.9</b>   | <b>10.1</b>  | <b>12.5</b>  | <b>15.5</b>  | <b>18.0</b>  | <b>19.8</b>  | <b>19.7</b>  | <b>17.7</b>  | <b>14.7</b>  | <b>11.8</b>  | <b>9.7</b>  |
|                                                                               |             | MCDB         | <b>10.0</b>  | <b>9.4</b>   | <b>11.1</b>  | <b>14.5</b>  | <b>18.0</b>  | <b>21.2</b>  | <b>22.8</b>  | <b>22.2</b>  | <b>19.7</b>  | <b>15.4</b>  | <b>12.2</b>  | <b>10.3</b> |
|                                                                               | 5%          | WB           | <b>8.9</b>   | <b>8.2</b>   | <b>9.2</b>   | <b>11.5</b>  | <b>14.7</b>  | <b>16.9</b>  | <b>18.9</b>  | <b>18.9</b>  | <b>16.9</b>  | <b>13.8</b>  | <b>11.1</b>  | <b>8.9</b>  |
|                                                                               |             | MCDB         | <b>9.4</b>   | <b>8.8</b>   | <b>10.2</b>  | <b>13.2</b>  | <b>17.1</b>  | <b>19.8</b>  | <b>21.8</b>  | <b>21.4</b>  | <b>18.8</b>  | <b>14.4</b>  | <b>11.6</b>  | <b>9.5</b>  |
|                                                                               | 10%         | WB           | <b>8.2</b>   | <b>7.7</b>   | <b>8.6</b>   | <b>10.5</b>  | <b>14.0</b>  | <b>16.0</b>  | <b>18.1</b>  | <b>18.2</b>  | <b>16.2</b>  | <b>12.7</b>  | <b>10.2</b>  | <b>8.1</b>  |
|                                                                               |             | MCDB         | <b>8.7</b>   | <b>8.3</b>   | <b>9.6</b>   | <b>12.0</b>  | <b>16.4</b>  | <b>18.6</b>  | <b>20.9</b>  | <b>20.7</b>  | <b>17.9</b>  | <b>13.5</b>  | <b>10.8</b>  | <b>8.8</b>  |
| (35) <b>Mean Daily Temperature Range</b>                                      | 5% DB       | MDBR         | <b>2.6</b>   | <b>2.9</b>   | <b>3.4</b>   | <b>4.0</b>   | <b>4.4</b>   | <b>4.5</b>   | <b>4.6</b>   | <b>4.3</b>   | <b>3.7</b>   | <b>3.0</b>   | <b>2.9</b>   | <b>2.6</b>  |
|                                                                               |             | MCDBR        | <b>3.3</b>   | <b>3.2</b>   | <b>4.3</b>   | <b>5.3</b>   | <b>5.8</b>   | <b>6.0</b>   | <b>5.9</b>   | <b>5.4</b>   | <b>4.6</b>   | <b>4.1</b>   | <b>3.8</b>   | <b>3.4</b>  |
|                                                                               |             | MCWBR        | <b>2.7</b>   | <b>2.5</b>   | <b>3.1</b>   | <b>3.6</b>   | <b>3.9</b>   | <b>3.9</b>   | <b>3.8</b>   | <b>3.6</b>   | <b>3.3</b>   | <b>3.4</b>   | <b>3.5</b>   | <b>2.8</b>  |
|                                                                               | 5% WB       | MCDBR        | <b>3.1</b>   | <b>3.0</b>   | <b>4.0</b>   | <b>5.0</b>   | <b>5.3</b>   | <b>5.4</b>   | <b>5.5</b>   | <b>4.9</b>   | <b>4.3</b>   | <b>3.8</b>   | <b>3.7</b>   | <b>3.4</b>  |
|                                                                               |             | MCWBR        | <b>2.7</b>   | <b>2.6</b>   | <b>3.1</b>   | <b>3.6</b>   | <b>3.7</b>   | <b>3.9</b>   | <b>3.9</b>   | <b>3.7</b>   | <b>3.4</b>   | <b>3.4</b>   | <b>3.4</b>   | <b>3.0</b>  |
| (40) <b>Clear-Sky Solar Irradiance</b>                                        | taub        | <b>0.315</b> | <b>0.307</b> | <b>0.328</b> | <b>0.346</b> | <b>0.367</b> | <b>0.360</b> | <b>0.348</b> | <b>0.350</b> | <b>0.334</b> | <b>0.332</b> | <b>0.314</b> | <b>0.308</b> |             |
|                                                                               | taud        | <b>2.499</b> | <b>2.568</b> | <b>2.520</b> | <b>2.475</b> | <b>2.434</b> | <b>2.469</b> | <b>2.548</b> | <b>2.548</b> | <b>2.582</b> | <b>2.559</b> | <b>2.546</b> | <b>2.486</b> |             |
|                                                                               | Ebn at Noon | <b>746</b>   | <b>857</b>   | <b>890</b>   | <b>904</b>   | <b>893</b>   | <b>899</b>   | <b>906</b>   | <b>890</b>   | <b>873</b>   | <b>811</b>   | <b>746</b>   | <b>702</b>   |             |
|                                                                               | Edh at Noon | <b>60</b>    | <b>70</b>    | <b>87</b>    | <b>100</b>   | <b>109</b>   | <b>106</b>   | <b>97</b>    | <b>92</b>    | <b>81</b>    | <b>70</b>    | <b>57</b>    | <b>54</b>    |             |
| (44) <b>All-Sky Solar Radiation</b>                                           | RadAvg      | <b>0.84</b>  | <b>1.73</b>  | <b>2.57</b>  | <b>4.00</b>  | <b>5.20</b>  | <b>5.60</b>  | <b>6.16</b>  | <b>5.25</b>  | <b>3.69</b>  | <b>2.01</b>  | <b>1.00</b>  | <b>0.66</b>  |             |
|                                                                               | RadStd      | <b>0.13</b>  | <b>0.27</b>  | <b>0.34</b>  | <b>0.41</b>  | <b>0.51</b>  | <b>0.64</b>  | <b>0.58</b>  | <b>0.43</b>  | <b>0.32</b>  | <b>0.30</b>  | <b>0.18</b>  | <b>0.10</b>  |             |

**Historical Trends**

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only            |       | N/A     | -0.57  | -1.42   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (61 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |